

Proportion & Rule of Three — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 · MCQ**[1]**

If $a : b :: c : d$, then:

(a) $ac = bd$ (b) $ad = bc$ (c) $a + d = b + c$ (d) $ab = cd$

Q2 · MCQ**[1]**

Which of these is a true proportion?

(a) $8 : 3 :: 24 : 6$ (b) $7 : 12 :: 12 : 7$ (c) $21 : 6 :: 35 : 10$ (d) $12 : 18 :: 28 : 12$

Q3 · FILL IN THE BLANK**[1]**

In $6 : 10 :: 18 : x$, the value of x is _____.

Q4 · MCQ**[1]**

If $3 : 4 :: x : 20$, then x equals:

(a) 12 (b) 15 (c) 16 (d) 24

Q5 · TRUE / FALSE**[1]**

To use the Rule of Three, both ratios must use the same units.

True False

Q6 · ASSERTION-REASON**[1]**

Assertion (A): In $150 : 90 :: 240 : x$, the value of x is 144.

Reason (R): When $a : b :: c : d$, then $ad = bc$.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7**[2]**

Check whether $24 : 8$ and $9 : 3$ are proportional, using cross multiplication.

Q8**[2]**

6 glasses of lemonade need 10 spoons of sugar. How many spoons are needed for 18 glasses of the same sweetness?

Q9**[2]**

Find x : $5 : 8 :: x : 24$.

Long answer (3 marks each)

Q10**[3]**

A car travels 90 km in 150 minutes. At the same speed, what distance will it cover in 4 hours?
(Convert the units first.)

Q11

[3]

For 120 students a cook makes 15 kg of rice. How much rice is needed for 80 students?

Q12

[3]

Is $8 : 3 :: 24 : 6$ a true proportion? If not, find the fourth term that would make $8 : 3 :: 24 : ?$ a true proportion.

Total: 21 marks · Check your work against the [answer key](#).